

MH3100 Real Analysis I

Tutorial 4 Solutions

Academic Year 2026/2027, Semester 1

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Overview

- **Lecture coverage:** Lectures 10–12 — series, the Cauchy Criterion for Series, the Alternating Series Test, the Comparison Test, the Absolute Convergence Test, absolute vs. conditional convergence, and rearrangements.
- **Questions:** 8 questions (Questions 1–5, Additional Questions A1–A3); solutions extend the official hints, which mostly give only the final answer.
- **Notation:** $\mathbb{N} = \{1, 2, 3, \dots\}$ ($0 \notin \mathbb{N}$ in this course); series $\sum a_n$ with partial sum $s_n := a_1 + \dots + a_n$; results cited by slide name.
- **Running lemmas:** geometric series, $\sum 1/n^2$, and divergence of $\sum 1/n$ are proved once in Question 1 and reused throughout.

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Question 1 (Convergence of three series)

Problem

Determine if each of the following series is convergent or divergent.

(a) $\sum \frac{1}{2^n + n}.$

(b) $\sum \frac{\sin(n)}{n^2}.$

(c) $1 - \frac{1}{2^2} + \frac{1}{3} - \frac{1}{4^2} + \frac{1}{5} - \frac{1}{6^2} + \frac{1}{7} - \frac{1}{8^2} + \cdots.$

(Tutorial 4, Question 1.)

Official hint. Answers only: (a) convergent; (b) convergent; (c) divergent. No method, no comparison series, and no explanation of the pattern in (c) are given.

Solution

Method 1: Comparison Test, with two running lemmas

Lemma A (Geometric series). For $r \in \mathbb{R}$ with $0 < r < 1$, the series $\sum r^n$ is convergent, with sum $\frac{r}{1-r}$. (Scratch work – not part of the written answer.) The n -th partial

sum telescopes: $(1-r)(r + r^2 + \cdots + r^n) = r - r^{n+1}$, so $s_n = \frac{r - r^{n+1}}{1-r}$. Since $0 < r < 1$, the sequence $\{r^{n+1}\}$ converges to 0 (a standard fact about geometric sequences with ratio in $(0, 1)$).

Proof. Let $0 < r < 1$. For every $n \in \mathbb{N}$, direct expansion gives $(1-r)\sum_{k=1}^n r^k = \sum_{k=1}^n r^k - \sum_{k=1}^n r^{k+1} = r - r^{n+1}$, so that $s_n = \frac{r - r^{n+1}}{1-r}$. Since $r^n \rightarrow 0$ as $n \rightarrow \infty$, the Algebraic Limit Theorem gives $s_n \rightarrow \frac{r-0}{1-r} = \frac{r}{1-r}$. Therefore $\{s_n\}$ converges, so the series $\sum r^n$ is convergent, with sum $\frac{r}{1-r}$. $\square$

Lemma B (The series $\sum 1/n^2$). The series $\sum 1/n^2$ is convergent. (Scratch work – not part of the written answer.) We bound each term by a telescoping term and use that increasing, bounded partial sums converge (Monotone Convergence Theorem; this is allowed here, since only Question 4 of this tutorial bans it).

Proof. For every $n \in \mathbb{N}$ with $n \geq 2$, $0 < \frac{1}{n^2} \leq \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}$, because $n^2 \geq n(n-1)$. Therefore, for every $N \in \mathbb{N}$ with $N \geq 2$, the N -th partial sum $t_N = \sum_{n=1}^N \frac{1}{n^2}$ satisfies

$$t_N = 1 + \sum_{n=2}^N \frac{1}{n^2} \leq 1 + \sum_{n=2}^N \left(\frac{1}{n-1} - \frac{1}{n} \right) = 1 + \left(1 - \frac{1}{N} \right) \leq 2,$$

using that the sum telescopes. Since the terms $1/n^2$ are positive, $\{t_N\}$ is increasing; and it is bounded above by 2. Hence, by the Monotone Convergence Theorem, $\{t_N\}$ converges, so $\sum 1/n^2$ is convergent. $\square$

Lemma C (Harmonic series diverges). *The series $\sum 1/n$ is divergent; in fact its partial sums are unbounded. (Scratch work – not part of the written answer.)* Oresme's classical grouping argument: group the terms between consecutive powers of 2 and bound each group below by $\frac{1}{2}$.

Proof. We show by induction on $m \in \mathbb{N}_0$ that $\sum_{n=1}^{2^m} \frac{1}{n} \geq 1 + \frac{m}{2}$. For $m = 0$, the sum is $1 \geq 1 + 0$. Assume the claim holds for some $m \in \mathbb{N}_0$. The block of terms with indices from $2^m + 1$ to 2^{m+1} has $2^{m+1} - 2^m = 2^m$ terms, and each such term is at least $\frac{1}{2^{m+1}}$ since its index is at most 2^{m+1} . Therefore

$$\sum_{n=1}^{2^{m+1}} \frac{1}{n} = \sum_{n=1}^{2^m} \frac{1}{n} + \sum_{n=2^m+1}^{2^{m+1}} \frac{1}{n} \geq \left(1 + \frac{m}{2}\right) + 2^m \cdot \frac{1}{2^{m+1}} = 1 + \frac{m+1}{2},$$

which completes the induction. Thus the subsequence $\{u_m\}$ of partial sums of $\sum 1/n$, where $u_m := \sum_{n=1}^{2^m} 1/n$, satisfies $u_m \geq 1 + m/2 \rightarrow \infty$, so $\{u_m\}$ is unbounded. Since the partial sums of $\sum 1/n$ are increasing (the terms are positive) and contain the unbounded subsequence $\{u_m\}$, the full sequence of partial sums is itself unbounded. If $\sum 1/n$ were convergent, its sequence of partial sums would be bounded, by the Boundedness of Convergent Sequences; since it is unbounded, $\sum 1/n$ is therefore divergent. $\square$

Part (a). *(Scratch work – not part of the written answer.)* $2^n + n \geq 2^n$ term by term, so $\frac{1}{2^n + n} \leq \frac{1}{2^n} = \left(\frac{1}{2}\right)^n$, and $\sum (1/2)^n$ converges by Lemma A.

Proof. For every $n \in \mathbb{N}$, $n \geq 1 > 0$, so $2^n + n > 2^n > 0$; taking reciprocals reverses the inequality, giving

$$0 < \frac{1}{2^n + n} \leq \frac{1}{2^n} = \left(\frac{1}{2}\right)^n \quad \text{for all } n \in \mathbb{N}.$$

By Lemma A with $r = \frac{1}{2} \in (0, 1)$, the series $\sum (1/2)^n$ is convergent. Since the terms of $\sum \frac{1}{2^n + n}$ are nonnegative and dominated termwise by those of the convergent series $\sum (1/2)^n$, the Comparison Test asserts that $\sum \frac{1}{2^n + n}$ is convergent.

(a) $\sum \frac{1}{2^n + n}$ is convergent.

$\square$

Part (b). *(Scratch work – not part of the written answer.)* $|\sin n| \leq 1$, so $|\sin(n)/n^2| \leq 1/n^2$; apply the Absolute Convergence Test using Lemma B.

Proof. For every $n \in \mathbb{N}$, $-1 \leq \sin(n) \leq 1$, so $|\sin(n)| \leq 1$, and hence

$$\left| \frac{\sin(n)}{n^2} \right| = \frac{|\sin(n)|}{n^2} \leq \frac{1}{n^2} \quad \text{for all } n \in \mathbb{N}.$$

The terms $1/n^2$ are nonnegative, and by Lemma B the series $\sum 1/n^2$ is convergent. Since $|\sin(n)/n^2| \leq 1/n^2$ for every n , the Comparison Test (applied to the nonnegative sequences $\{|\sin(n)/n^2|\}$ and $\{1/n^2\}$) shows that $\sum |\sin(n)/n^2|$ is convergent; that is,

$\sum \sin(n)/n^2$ is absolutely convergent. By the Absolute Convergence Test, $\sum \sin(n)/n^2$ is therefore convergent.

(b) $\sum \frac{\sin(n)}{n^2}$ is convergent (in fact, absolutely).

□

Part (c). (*Scratch work – not part of the written answer.*) Read off the pattern of terms first. Writing the series as $\sum_{n=1}^{\infty} a_n$, the odd-indexed terms ($n = 1, 3, 5, \dots$) are $a_{2k-1} = 1/(2k-1)$ and the even-indexed terms ($n = 2, 4, 6, \dots$) are $a_{2k} = -1/(2k)^2$, for $k = 1, 2, 3, \dots$: check $k = 1$: $a_1 = 1$, $a_2 = -1/4$; $k = 2$: $a_3 = 1/3$, $a_4 = -1/16$; $k = 3$: $a_5 = 1/5$, $a_6 = -1/36$; $k = 4$: $a_7 = 1/7$, $a_8 = -1/64$ – all match the printed series. This is *not* a standard alternating series, since the two “halves” have different magnitudes ($1/(2k-1)$ versus $1/(2k)^2$), so the Alternating Series Test does not apply. Split the even-position partial sum into a divergent positive piece and a convergent negative piece.

Proof. For $k \in \mathbb{N}$, write $b_k := \frac{1}{2k-1}$ (the positive, odd-position terms) and $c_k := \frac{1}{(2k)^2}$ (the magnitudes of the negative, even-position terms), so that the given series is $\sum_{k=1}^{\infty} (b_k - c_k)$ in the sense that its $2m$ -th partial sum is $S_{2m} = \sum_{k=1}^m b_k - \sum_{k=1}^m c_k$.

First, $\sum c_k$ is convergent: for every $k \in \mathbb{N}$, $0 \leq c_k = \frac{1}{4k^2} \leq \frac{1}{k^2}$, and $\sum 1/k^2$ is convergent by Lemma B, so by the Comparison Test $\sum c_k$ is convergent, say to $C \in \mathbb{R}$; being convergent, its partial sums $\sum_{k=1}^m c_k$ form a bounded sequence, so there is $C' > 0$ with $\sum_{k=1}^m c_k \leq C'$ for all $m \in \mathbb{N}$.

Second, $\sum b_k$ is divergent with partial sums tending to infinity: for every $k \in \mathbb{N}$, $2k-1 < 2k$, so $b_k = \frac{1}{2k-1} \geq \frac{1}{2k}$. Hence for every $m \in \mathbb{N}$, $\sum_{k=1}^m b_k \geq \sum_{k=1}^m \frac{1}{2k} = \frac{1}{2} \sum_{k=1}^m \frac{1}{k}$. By Lemma C, $\sum_{k=1}^m 1/k \rightarrow \infty$ as $m \rightarrow \infty$ (its partial sums are unbounded above); therefore $\sum_{k=1}^m b_k$ is unbounded above as well.

Combining the two facts, for every $m \in \mathbb{N}$,

$$S_{2m} = \sum_{k=1}^m b_k - \sum_{k=1}^m c_k \geq \sum_{k=1}^m b_k - C',$$

and since $\sum_{k=1}^m b_k$ is unbounded above, so is S_{2m} (as $m \rightarrow \infty$). Thus the subsequence $\{S_{2m}\}$ of the sequence of partial sums $\{S_N\}$ of the given series is unbounded. If $\{S_N\}$ were convergent, then, by the theorem on Subsequential Limits, every subsequence – in particular $\{S_{2m}\}$ – would converge to the same limit, and would therefore be bounded, by the Boundedness of Convergent Sequences. Since $\{S_{2m}\}$ is unbounded, this is a contradiction. Therefore $\{S_N\}$ diverges, so the given series is divergent.

(c) $1 - \frac{1}{2^2} + \frac{1}{3} - \frac{1}{4^2} + \frac{1}{5} - \frac{1}{6^2} + \dots$ is divergent.

□

Beyond the hints. The official hints give only the three answers. All three comparison series (Lemmas A–C), the termwise inequalities, and – for part (c) – the identification of the term pattern and the splitting argument via Subsequential Limits are supplied here in full; none of this appears in the official hint.

Common slip. The Comparison Test requires the *smaller* sequence to be *nonnegative* for all n (not merely eventually, and not merely that the series converges); in part (c) it is tempting to try to apply the Alternating Series Test directly to the printed series, but its terms do not have a single common decreasing envelope $\{b_n\} \rightarrow 0$, since the odd- and even-position terms decay at different rates – this is exactly why the series is not treated as one alternating series here.

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Question 2 (True/false with examples)

Problem

Determine if each of the following statements is true.

- (a) There are two divergent series $\sum a_n$ and $\sum b_n$ where $\sum a_n b_n$ converges.
- (b) There is a convergent series $\sum a_n$ and a bounded sequence $\{b_n\}$ such that $\sum a_n b_n$ diverges.
- (c) There are two sequences $\{a_n\}$ and $\{b_n\}$ where $\sum a_n$ and $\sum(a_n + b_n)$ both converge but $\sum b_n$ diverges.
- (d) There is a sequence $\{a_n\}$ satisfying $0 \leq a_n \leq \frac{1}{n}$ where $\sum(-1)^n a_n$ diverges.

(Tutorial 4, Question 2; part (a) is also Sample Midterm Question 2(a).)

Official hint. Answers only: (a) True; (b) True; (c) False; (d) True. No justification, examples, or proofs are given.

Solution

Method 1: One explicit, verified example (or proof) per part

Part (a). (Scratch work – not part of the written answer.) Both $\sum a_n, \sum b_n$ must diverge individually, yet their termwise product series converges; take $a_n = b_n = 1/n$, so $a_n b_n = 1/n^2$.

Verification. Let $a_n = b_n = \frac{1}{n}$ for all $n \in \mathbb{N}$. By Lemma C (Question 1), $\sum a_n = \sum 1/n$ is divergent, and likewise $\sum b_n = \sum 1/n$ is divergent. However, $a_n b_n = \frac{1}{n^2}$ for every n , and by Lemma B (Question 1), $\sum a_n b_n = \sum 1/n^2$ is convergent. Therefore two divergent series $\sum a_n, \sum b_n$ exist with $\sum a_n b_n$ convergent.

(a) True, e.g. $a_n = b_n = \frac{1}{n}$.

Part (b). (Scratch work – not part of the written answer.) Need $\sum a_n$ convergent, $\{b_n\}$ bounded, $\sum a_n b_n$ divergent; try to turn a conditionally convergent alternating series into the divergent harmonic series by multiplying by another copy of $(-1)^n$.

Verification. Let $a_n = \frac{(-1)^n}{n}$ and $b_n = (-1)^n$ for all $n \in \mathbb{N}$. The sequence $\{1/n\}$ is decreasing and converges to 0, so by the Alternating Series Test (applied to $-\sum a_n = \sum(-1)^{n-1}(1/n)$, hence also to $\sum a_n$ by the Algebraic Limit Theorem), $\sum a_n$ is convergent. Also $|b_n| = 1$ for all n , so $\{b_n\}$ is bounded. But for every $n \in \mathbb{N}$,

$$a_n b_n = \frac{(-1)^n}{n} \cdot (-1)^n = \frac{((-1)^n)^2}{n} = \frac{1}{n},$$

and $\sum a_n b_n = \sum 1/n$ is divergent by Lemma C. Therefore such a convergent series and bounded sequence exist.

(b) True, e.g. $a_n = \frac{(-1)^n}{n}$, $b_n = (-1)^n$.

Part (c). (*Scratch work – not part of the written answer.*) If $\sum a_n$ and $\sum(a_n + b_n)$ both converge, write b_n as the difference of their terms and apply the Algebraic Limit Theorem for Series – this forces $\sum b_n$ to converge, so no such divergent $\sum b_n$ can exist.

Proof (statement is false). Suppose $\{a_n\}$ and $\{b_n\}$ are sequences for which $\sum a_n$ converges, say to $s \in \mathbb{R}$, and $\sum(a_n + b_n)$ converges, say to $t \in \mathbb{R}$. For every $n \in \mathbb{N}$, $b_n = (a_n + b_n) - a_n$. Since $\sum(a_n + b_n)$ converges to t and $\sum a_n$ converges to s , the Algebraic Limit Theorem for Series (applied with $c = -1$ to $\sum a_n$, then added to $\sum(a_n + b_n)$) gives that $\sum b_n = \sum((a_n + b_n) - a_n)$ converges, to $t - s$. Hence it is impossible for $\sum a_n$ and $\sum(a_n + b_n)$ to both converge while $\sum b_n$ diverges.

$$(c) \text{ False: } \sum a_n, \sum(a_n + b_n) \text{ convergent} \implies \sum b_n \text{ convergent.}$$

□

Part (d). (*Scratch work – not part of the written answer.*) The hypothesis $0 \leq a_n \leq 1/n$ alone does not force $a_n \rightarrow 0$ monotonically enough for the Alternating Series Test; kill off every other term so the surviving terms form (half of) the divergent harmonic series with a constant sign.

Verification. Define $a_n := \begin{cases} 1/n & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$ for $n \in \mathbb{N}$. Then $0 \leq a_n \leq 1/n$ holds

for every n : for even n , $a_n = 1/n$; for odd n , $a_n = 0 \leq 1/n$. Consider the partial sums $T_N := \sum_{n=1}^N (-1)^n a_n$. Since $a_n = 0$ for odd n , only even indices contribute, and $(-1)^n = 1$ for even n , so for every $m \in \mathbb{N}$,

$$\begin{aligned} T_{2m} &= \sum_{k=1}^m (-1)^{2k} a_{2k} = \sum_{k=1}^m \frac{1}{2k} = \frac{1}{2} \sum_{k=1}^m \frac{1}{k}, \\ T_{2m+1} &= T_{2m} + (-1)^{2m+1} a_{2m+1} = T_{2m} + 0 = T_{2m}. \end{aligned}$$

By Lemma C, $\sum_{k=1}^m 1/k \rightarrow \infty$, so $T_{2m} \rightarrow \infty$; since $T_{2m+1} = T_{2m}$ for every m , the whole sequence $\{T_N\}$ is unbounded. By the Boundedness of Convergent Sequences (contrapositive), $\{T_N\}$ diverges, i.e. $\sum (-1)^n a_n$ is divergent.

$$(d) \text{ True, e.g. } a_n = 1/n \text{ for even } n, a_n = 0 \text{ for odd } n.$$

Beyond the hints. The official hint supplies only the four True/False answers. Every example (and, for part (c), the full proof of falsity) is constructed and verified above; the digest of the official materials notes that a naive first attempt at (a) and (b) (e.g. $a_n = b_n = (-1)^n$ for (a)) does *not* work, since $a_n b_n = 1$ for all n still diverges – the examples chosen here were checked to actually satisfy every hypothesis.

Common slip. In part (d), trying $a_n = 1/n$ for *all* n gives the alternating harmonic series $\sum (-1)^n a_n$, which *converges* by the Alternating Series Test (since $1/n$ is decreasing to 0) – this is why the counterexample must break the “decreasing” hypothesis of that test, not merely satisfy $0 \leq a_n \leq 1/n$.

Question 3 (Proofs or counterexamples)

Problem

For each of the following statements, provide either a short proof if it is true or a counterexample if it is false.

- (a) If a series $\sum a_n$ converges absolutely, then the series $\sum a_n^2$ also converges absolutely.
- (b) If a series $\sum a_n$ converges and a sequence $\{b_n\}$ converges, then the series $\sum a_n b_n$ converges.
- (c) If a series $\sum a_n$ converges conditionally, then the series $\sum n^2 a_n$ diverges.

(Tutorial 4, Question 3.)

Official hint. (a) Apply the Comparison Test using the Terms of Convergent Series, the Boundedness of Convergent Sequences, and the Algebraic Limit Theorem. (b) Find a conditionally convergent series written as $\sum a_n b_n$ with $\sum a_n$ conditionally convergent and $\{b_n\}$ convergent. (c) Prove the contrapositive, applying the Terms of Convergent Series to $\sum n^2 a_n$, then the Comparison Test with a specific absolutely convergent series. No inequality chains, comparison series, or worked examples are given.

Solution

Method 1: Direct comparison (a, c) and a conditionally convergent product (b)

Part (a) – statement is true. (*Scratch work – not part of the written answer.*) Since $\sum |a_n|$ converges, $\{|a_n|\} \rightarrow 0$, hence (Boundedness of Convergent Sequences) $\{|a_n|\}$ is bounded by some M for *all* n – not just eventually – which turns $a_n^2 \leq M|a_n|$ into a global (not merely eventual) comparison, avoiding any issue with finitely many exceptional indices.

Proof. Suppose $\sum a_n$ converges absolutely, i.e. $\sum |a_n|$ converges. By the theorem on the Terms of Convergent Series, applied to the series $\sum |a_n|$, the sequence $\{|a_n|\}$ converges to 0. By the Boundedness of Convergent Sequences, the convergent sequence $\{|a_n|\}$ is bounded; that is, there is some $M > 0$ such that $|a_n| \leq M$ for every $n \in \mathbb{N}$. Consequently, for every $n \in \mathbb{N}$,

$$0 \leq a_n^2 = |a_n| \cdot |a_n| \leq M \cdot |a_n|.$$

By the Algebraic Limit Theorem for Series, $\sum M|a_n|$ converges (it is a constant multiple of the convergent series $\sum |a_n|$). Since $0 \leq a_n^2 \leq M|a_n|$ for every $n \in \mathbb{N}$, the Comparison Test shows that $\sum a_n^2$ is convergent. Moreover $a_n^2 \geq 0$ for every n , so $a_n^2 = |a_n^2|$, and hence $\sum |a_n^2| = \sum a_n^2$ is convergent; that is, $\sum a_n^2$ converges absolutely.

(a) True: $\sum a_n$ absolutely convergent $\implies \sum a_n^2$ absolutely convergent.

□

Part (b) – statement is false. (*Scratch work – not part of the written answer.*) Need a convergent $\sum a_n$, a convergent $\{b_n\}$, with $\sum a_n b_n$ divergent. Square a conditionally convergent alternating series against itself: signs cancel to $+1$, magnitudes multiply to $1/n$.

Counterexample. Let $a_n = \frac{(-1)^n}{\sqrt{n}}$ and $b_n = \frac{(-1)^n}{\sqrt{n}}$ for $n \in \mathbb{N}$. The sequence $\{1/\sqrt{n}\}$ is decreasing and converges to 0 (by the Archimedean Property), so by the Alternating Series Test, $\sum a_n$ converges. The sequence $\{b_n\}$ satisfies $|b_n| = 1/\sqrt{n} \rightarrow 0$, so $\{b_n\}$ converges (to 0). However, for every $n \in \mathbb{N}$,

$$a_n b_n = \frac{(-1)^n}{\sqrt{n}} \cdot \frac{(-1)^n}{\sqrt{n}} = \frac{((-1)^n)^2}{n} = \frac{1}{n},$$

and $\sum a_n b_n = \sum 1/n$ is divergent, by Lemma C (Question 1). Therefore $\sum a_n$ converges and $\{b_n\}$ converges, yet $\sum a_n b_n$ diverges, so the statement is false.

(b) False, e.g. $a_n = b_n = \frac{(-1)^n}{\sqrt{n}}$.

Part (c) – statement is true. (*Scratch work – not part of the written answer.*) Prove the contrapositive: if $\sum n^2 a_n$ converges, show $\sum a_n$ converges *absolutely*, which rules out conditional convergence. Use Terms of Convergent Series on $\sum n^2 a_n$ to bound $|a_n|$ by M/n^2 , then Comparison Test with Lemma B.

Proof. We prove the contrapositive: if $\sum n^2 a_n$ converges, then $\sum a_n$ does not converge conditionally. Suppose $\sum n^2 a_n$ converges. By the theorem on the Terms of Convergent Series, applied to $\sum n^2 a_n$, the sequence $\{n^2 a_n\}$ converges to 0. By the Boundedness of Convergent Sequences, $\{n^2 a_n\}$ is bounded: there is $M > 0$ with $|n^2 a_n| \leq M$ for every $n \in \mathbb{N}$, i.e.

$$0 \leq |a_n| \leq \frac{M}{n^2} \quad \text{for all } n \in \mathbb{N}.$$

By Lemma B, $\sum 1/n^2$ converges, so by the Algebraic Limit Theorem for Series, $\sum M/n^2$ converges. Since $0 \leq |a_n| \leq M/n^2$ for every n , the Comparison Test gives that $\sum |a_n|$ converges; that is, $\sum a_n$ converges absolutely. In particular, if $\sum a_n$ converges at all, it cannot converge *conditionally* (conditional convergence means convergent but not absolutely convergent, by definition). This proves the contrapositive; therefore, if $\sum a_n$ converges conditionally, then $\sum n^2 a_n$ diverges.

(c) True: $\sum a_n$ conditionally convergent $\implies \sum n^2 a_n$ divergent.

□

Beyond the hints. The official hint names the theorems to combine but never writes the inequality chain or a concrete pair of series; both are supplied here, with every hypothesis (decreasing, bounded, nonnegative comparison) checked explicitly, and the counterexample in (b) is verified by direct computation rather than merely asserted.

Common slip. In part (c), it is tempting to argue directly (“ $\sum a_n$ conditionally convergent $\implies$ terms don’t decay fast $\implies \sum n^2 a_n$ diverges”), but this is not a valid deduction on its own; the contrapositive route is needed because convergence of $\sum n^2 a_n$ is exactly the hypothesis that lets us extract a decay rate $|a_n| = O(1/n^2)$, whereas conditional convergence of $\sum a_n$ alone says nothing quantitative about how fast $a_n \rightarrow 0$.

Question 4 (Cauchy partial sums without the named tests)

Problem

Prove that if the series $\sum |a_n|$ has a Cauchy sequence of partial sums and the sequence $\{b_n\}$ is bounded, then the series $\sum a_n b_n$ has a Cauchy sequence of partial sums. **Do not use the Cauchy Criterion, the Comparison Test, or the Absolute Convergence Test.** (*Tutorial 4, Question 4.*)

Official hint. (a) Write down the definitions for the sequences of partial sums of both $\sum |a_n|$ and $\sum a_n b_n$ to be Cauchy, and use the boundedness of $\{b_n\}$ to compare them. (b) Follow the proof of the Absolute Convergence Test via the Cauchy Criterion. No explicit inequality or ε -bookkeeping is given.

Solution

Method 1: Re-derive the comparison estimate from the bare Cauchy definitions

Idea. (*Scratch work – not part of the written answer.*) We are barred from citing the Cauchy Criterion, Comparison Test, or Absolute Convergence Test by name, so the estimate that normally proves those theorems must be re-derived here from the raw ε - N definition of “Cauchy sequence of partial sums”, using only the triangle inequality and the definition of a bounded sequence. Let $\{s_n\}$ denote the partial sums of $\sum |a_n|$ and $\{t_n\}$ the partial sums of $\sum a_n b_n$. The key algebraic step is $|t_n - t_m| = |\sum_{k=m+1}^n a_k b_k| \leq M \sum_{k=m+1}^n |a_k| = M |s_n - s_m|$, where M bounds $\{b_n\}$; this lets an ε for $\{s_n\}$ produce an ε for $\{t_n\}$ after rescaling by M .

Proof. Since $\{b_n\}$ is bounded, there is $M > 0$ such that $|b_n| \leq M$ for every $n \in \mathbb{N}$. Let $s_n := |a_1| + |a_2| + \cdots + |a_n|$ and $t_n := a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$ denote the n -th partial sums of $\sum |a_n|$ and $\sum a_n b_n$ respectively.

Let $\varepsilon > 0$ be given. Since $\{s_n\}$ is a Cauchy sequence of partial sums, applying the definition of a Cauchy sequence with $\varepsilon' := \frac{\varepsilon}{M+1} > 0$, there exists $N \in \mathbb{N}$ such that for all $n, m \in \mathbb{N}$,

$$n > m \geq N \implies |s_n - s_m| = |a_{m+1}| + |a_{m+2}| + \cdots + |a_n| < \varepsilon'.$$

(The absolute value on $s_n - s_m$ may be dropped because $s_n \geq s_m$, as the terms $|a_k|$ are nonnegative.)

Fix $n, m \in \mathbb{N}$ with $n > m \geq N$. Then

$$\begin{aligned} |t_n - t_m| &= \left| \sum_{k=m+1}^n a_k b_k \right| \leq \sum_{k=m+1}^n |a_k| |b_k| \leq M \sum_{k=m+1}^n |a_k| \\ &= M(|s_n - s_m|) < M\varepsilon' = \frac{M\varepsilon}{M+1} < \varepsilon, \end{aligned}$$

where the first inequality is the triangle inequality applied repeatedly to the finite sum, the second uses $|b_k| \leq M$ for every k , and the last two use the choice of N and $\frac{M}{M+1} < 1$.

Therefore, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that for all $n, m \in \mathbb{N}$, $n > m \geq N \implies |t_n - t_m| < \varepsilon$. This is precisely the definition for the sequence of partial sums $\{t_n\}$ of $\sum a_n b_n$ to be Cauchy.

$$\boxed{\sum |a_n| \text{ Cauchy partial sums, } \{b_n\} \text{ bounded} \implies \sum a_n b_n \text{ has Cauchy partial sums.}}$$

□

Beyond the hints. The official hint tells the student *which* definitions to write down and *which* earlier proof to imitate, but never writes the inequality $|\sum_{k=m+1}^n a_k b_k| \leq M \sum_{k=m+1}^n |a_k|$ itself, nor states how to convert an ε for one Cauchy sequence into an ε for the other (the $\varepsilon' := \varepsilon/(M+1)$ rescaling, chosen to sidestep the case $M=0$ cleanly). Both are supplied above.

Common slip. It is tempting to write “by the Comparison Test” or “by the Absolute Convergence Test” at the key inequality step, since the estimate is literally the proof of those theorems – but the question explicitly forbids citing them by name; the estimate must instead be exhibited directly from the ε – N definitions, as done above. A second common slip is forgetting that $\{b_n\}$ bounded gives a single M that works for *all* k simultaneously (not depending on n, m), which is exactly what allows M to be pulled out of the sum.

Question 5 (Regrouping preserves the sum)

Problem

Given a series $\sum a_n$ and a strictly increasing sequence of natural numbers $\{n_k\}$, the regrouping of the terms

$$(a_1 + a_2 + \cdots + a_{n_1}) + (a_{n_1+1} + a_{n_1+2} + \cdots + a_{n_2}) + (a_{n_2+1} + a_{n_2+2} + \cdots + a_{n_3}) + \cdots$$

leads to a series $\sum b_k$ with the terms

$$b_k := a_{n_{k-1}+1} + a_{n_{k-1}+2} + \cdots + a_{n_k} \quad \forall k \in \mathbb{N},$$

where n_0 denotes 0. Prove that if $\sum a_n$ sums to s , then $\sum b_k$ sums to s . (*Tutorial 4, Question 5.*)

Official hint. “Compare the sequence of partial sums of $\sum b_k$ with that of $\sum a_n$.” No further detail is given.

Solution

Method 1: Partial sums of $\sum b_k$ are a subsequence of those of $\sum a_n$

Idea. (*Scratch work – not part of the written answer.*) The k -th partial sum of $\sum b_k$ telescopes down to s_{n_k} , the n_k -th partial sum of $\sum a_n$; since $\{n_k\}$ is strictly increasing, this exhibits the partial sums of $\sum b_k$ as exactly a subsequence of the partial sums of $\sum a_n$, and subsequences of a convergent sequence converge to the same limit.

Proof. Let $\{s_n\}$ denote the sequence of partial sums of $\sum a_n$, i.e. $s_n := a_1 + a_2 + \cdots + a_n$ for $n \in \mathbb{N}$ (and $s_0 := 0$), and let $\{\sigma_k\}$ denote the sequence of partial sums of $\sum b_k$, i.e. $\sigma_k := b_1 + b_2 + \cdots + b_k$ for $k \in \mathbb{N}$.

We first show that $\sigma_k = s_{n_k}$ for every $k \in \mathbb{N}$. By definition,

$$\sigma_k = \sum_{j=1}^k b_j = \sum_{j=1}^k (a_{n_{j-1}+1} + a_{n_{j-1}+2} + \cdots + a_{n_j}) = \sum_{i=n_0+1}^{n_k} a_i,$$

since the k blocks $\{n_{j-1} + 1, \dots, n_j\}$ for $j = 1, \dots, k$ partition the index set $\{n_0 + 1, \dots, n_k\} = \{1, \dots, n_k\}$ (using $n_0 = 0$) into consecutive, non-overlapping runs. Therefore $\sigma_k = \sum_{i=1}^{n_k} a_i = s_{n_k}$ for every $k \in \mathbb{N}$.

Since $\{n_k\}$ is a strictly increasing sequence of natural numbers, the map $k \mapsto n_k$ is exactly the index selection defining a subsequence: $\{s_{n_k}\}$ is a subsequence of $\{s_n\}$. By hypothesis, $\sum a_n$ sums to s , i.e. $\{s_n\}$ converges to s . By the theorem on Subsequential Limits, every subsequence of a convergent sequence converges to the same limit; hence $\{s_{n_k}\}$ converges to s . Since $\sigma_k = s_{n_k}$ for every k , the sequence $\{\sigma_k\}$ converges to s as well. By definition, this means $\sum b_k$ is convergent and sums to s .

$$\sum a_n \text{ sums to } s \implies \sum b_k \text{ sums to } s.$$

□

Beyond the hints. The official hint states only “compare the partial sums”. The precise identity $\sigma_k = s_{n_k}$ (proved above from the block-partition structure of the index set), the observation that this identifies $\{\sigma_k\}$ as a genuine subsequence of $\{s_n\}$, and the citation of the Subsequential Limits theorem to close the argument are all supplied here; none of this is in the hint.

Common slip. It is important that the index blocks $\{n_{j-1} + 1, \dots, n_j\}$ are consecutive and non-overlapping and exhaust $\{1, \dots, n_k\}$ exactly – this uses both that $\{n_k\}$ is strictly increasing (so no block is empty or reversed) and that $n_0 = 0$ (so the first block starts at index 1); without strict monotonicity, σ_k would not equal a single partial sum s_{n_k} of the original series.

Unofficial – QRS@NTU – qrs@ntu.org

Question A1 (Abel's Test)

Problem

Use the Cauchy Criterion for Series to prove that if the series $\sum a_n$ is convergent and the sequence $\{b_n\}$ is nonnegative and decreasing, then the series $\sum a_n b_n$ is convergent. **Do not use the Monotone Convergence Theorem or the Algebraic Limit Theorem for Series.** [Note: This is known as Abel's Test.] (Tutorial 4, Additional Question 1.)

Official hint. (a) Apply the Cauchy Criterion to $\sum a_n$. (b) Abel's Lemma: for $n, m \in \mathbb{N}$ with $m > n$, $\sum_{k=n+1}^m a_k b_k = s_m b_m - s_n b_{n+1} + \sum_{k=n+1}^{m-1} s_k (b_k - b_{k+1})$, where s_k is the k -th partial sum of $\sum a_n$. (c) "What happens to Abel's Lemma when $\{s_n\}$ is constant?" Neither the proof of Abel's Lemma, the bounding argument, nor the purpose of (c) is given.

Solution

Method 1: Abel's Lemma (summation by parts), then a Cauchy-Criterion estimate

Idea. (Scratch work – not part of the written answer.) First derive Abel's Lemma from scratch by writing $a_k = s_k - s_{k-1}$ and re-indexing. Hint (c) asks what happens when $\{s_n\}$ is constant: if $s_k = s_n$ for all $k \geq n$ (only possible when $a_k = 0$ eventually), the right-hand side of Abel's Lemma collapses to 0, which is consistent since then $a_k = 0$ for $k > n$ too. This suggests working with the *shifted* sequence $\sigma_k := s_k - s_n$ (so $\sigma_n = 0$, mimicking the "constant s_n " case at the base point n) rather than s_k itself: applying Abel's Lemma to σ removes the s_n -dependent boundary term entirely and produces a bound purely in terms of *tail* sums of $\sum a_n$, which the Cauchy Criterion makes small – with no need to ever establish $b_n \rightarrow 0$ (which is not assumed, and which the ban on the Monotone Convergence Theorem prevents us from deriving from "nonnegative and decreasing" alone).

Proof. *Step 1: Abel's Lemma.* Let $n, m \in \mathbb{N}$ with $m > n$. Since $a_k = s_k - s_{k-1}$ for every $k \geq 2$ (in particular for $k = n+1, \dots, m$, as $n \geq 1$),

$$\sum_{k=n+1}^m a_k b_k = \sum_{k=n+1}^m (s_k - s_{k-1}) b_k = \sum_{k=n+1}^m s_k b_k - \sum_{k=n+1}^m s_{k-1} b_k.$$

In the second sum, substitute $j = k - 1$, so j runs from n to $m - 1$ and the sum becomes $\sum_{j=n}^{m-1} s_j b_{j+1}$. Therefore

$$\begin{aligned} \sum_{k=n+1}^m a_k b_k &= \left[\sum_{k=n+1}^{m-1} s_k b_k + s_m b_m \right] - \left[s_n b_{n+1} + \sum_{k=n+1}^{m-1} s_k b_{k+1} \right] \\ &= s_m b_m - s_n b_{n+1} + \sum_{k=n+1}^{m-1} s_k (b_k - b_{k+1}), \end{aligned}$$

which is precisely Abel's Lemma.

Step 2: A tail form of Abel's Lemma. Fix $n \in \mathbb{N}$ and, for $k \geq n$, let $\sigma_k := s_k - s_n$ (so $\sigma_n = 0$). For $k \geq n+1$, $\sigma_k - \sigma_{k-1} = (s_k - s_n) - (s_{k-1} - s_n) = s_k - s_{k-1} = a_k$, so

$\{\sigma_k\}$ plays exactly the role of “the partial sums of $\sum a_n$, restarted at $\sigma_n = 0$ ”. Repeating the computation of Step 1 verbatim with σ in place of s (valid, since Step 1 used only $a_k = (\text{partial sum})_k - (\text{partial sum})_{k-1}$) gives, for $m > n$,

$$\sum_{k=n+1}^m a_k b_k = \sigma_m b_m - \sigma_n b_{n+1} + \sum_{k=n+1}^{m-1} \sigma_k (b_k - b_{k+1}) = (s_m - s_n) b_m + \sum_{k=n+1}^{m-1} (s_k - s_n) (b_k - b_{k+1}),$$

using $\sigma_n = 0$ and $\sigma_j = s_j - s_n$ for $j \geq n$.

Step 3: Global bound on $\{b_n\}$. Since $\{b_n\}$ is decreasing, $b_1 \geq b_2 \geq b_3 \geq \dots$, so $b_n \leq b_1$ for every $n \in \mathbb{N}$; together with $b_n \geq 0$, this gives $0 \leq b_n \leq b_1$ for all n – directly from the definition of “decreasing”, with no need for the Monotone Convergence Theorem.

Step 4: Cauchy estimate. Let $\varepsilon > 0$ be given, and set $\varepsilon' := \frac{\varepsilon}{b_1 + 1} > 0$. Since $\sum a_n$ is convergent, the Cauchy Criterion for Series gives $N \in \mathbb{N}$ such that for all $j, k \in \mathbb{N}$, $k \geq j \geq N \implies |s_k - s_j| < \varepsilon'$ (this includes $k = j$, trivially). Fix $n, m \in \mathbb{N}$ with $m > n \geq N$. By Step 2,

$$\left| \sum_{k=n+1}^m a_k b_k \right| \leq |s_m - s_n| b_m + \sum_{k=n+1}^{m-1} |s_k - s_n| (b_k - b_{k+1}),$$

using $b_m \geq 0$ and $b_k - b_{k+1} \geq 0$ (as $\{b_n\}$ is decreasing) to drop the absolute values on those two nonnegative factors. Since $n \geq N$, both $|s_m - s_n| < \varepsilon'$ and, for every k with $n+1 \leq k \leq m-1$ (so $k \geq n \geq N$), $|s_k - s_n| < \varepsilon'$. Therefore

$$\left| \sum_{k=n+1}^m a_k b_k \right| < \varepsilon' b_m + \varepsilon' \sum_{k=n+1}^{m-1} (b_k - b_{k+1}) = \varepsilon' b_m + \varepsilon' (b_{n+1} - b_m) = \varepsilon' b_{n+1} \leq \varepsilon' b_1 < \varepsilon,$$

where the middle sum telescopes to $b_{n+1} - b_m$, and the last two steps use $b_{n+1} \leq b_1$ (Step 3) and $\varepsilon'(b_1 + 1) = \varepsilon$, so $\varepsilon' b_1 < \varepsilon'(b_1 + 1) = \varepsilon$.

Thus, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that for all $n, m \in \mathbb{N}$, $m > n \geq N \implies \left| \sum_{k=n+1}^m a_k b_k \right| < \varepsilon$; this is the Cauchy Criterion condition for the partial sums of $\sum a_n b_n$. By the Cauchy Criterion for Series, $\sum a_n b_n$ is convergent.

$$\sum a_n \text{ convergent, } \{b_n\} \text{ nonnegative and decreasing } \implies \sum a_n b_n \text{ convergent.}$$

□

Beyond the hints. The official hint states Abel’s Lemma but never proves it, never explains hint (c), and never carries out the bounding argument. All three are supplied above; in particular, note that the proof *never* uses or needs $b_n \rightarrow 0$ – only that $\{b_n\}$ is nonnegative, decreasing (hence bounded by b_1), which is the correct generality for Abel’s Test (the sequence $\{b_n\}$ need not tend to 0; e.g. $b_n \equiv 1$ recovers the trivial case $\sum a_n b_n = \sum a_n$).

Common slip. Bounding $|s_k|$ by a single global constant M (as in the Comparison-Test-style estimates of Questions 1, 3, 4) and using $|\sum a_k b_k| \leq 2M b_{n+1}$ only shows convergence when $b_{n+1} \rightarrow 0$, which is *not* assumed here; the fix is to work with the tail difference $s_k - s_n$ (Step 2) rather than s_k itself, so that smallness comes from the Cauchy Criterion applied to $\sum a_n$ ’s tail, not from $\{b_n\} \rightarrow 0$.

Question A2 (Infinite products)

Problem

A close relative of the infinite series is the *infinite product*, which is a formal expression of the form

$$\prod b_n := \prod_{n=1}^{\infty} b_n := b_1 b_2 b_3 \cdots,$$

where $\{b_n\}$ is a sequence of real numbers. This infinite product is understood in terms of its sequence of *partial products* $\{p_n\}$, where, for each $n \in \mathbb{N}$, $p_n := b_1 b_2 b_3 \cdots b_n$. An infinite product is said to be *convergent* if its sequence of partial products is convergent.

- (a) Use the Cauchy Criterion to prove that if $\{b_n\}$ and $\{c_n\}$ are sequences for which $1 \leq b_n \leq c_n$ for all $n \in \mathbb{N}$ and $\prod c_n$ is convergent, then $\prod b_n$ is also convergent. **Do not use the Monotone Convergence Theorem.**

For the rest of this question, we consider the special case of infinite products of the form

$$\prod (1 + a_n) = (1 + a_1)(1 + a_2)(1 + a_3) \cdots,$$

where the sequence $\{a_n\}$ is nonnegative.

- (b) Find an explicit formula for the partial products in the case where $a_n = \frac{1}{n}$. Is the infinite product convergent?
- (c) Use the Cauchy Criterion to prove that the infinite product $\prod (1 + a_n)$ is convergent if and only if the series $\sum a_n$ is convergent. **Do not use the Monotone Convergence Theorem.**

(Tutorial 4, Additional Question 2.)

Official hint. (a) Follow the proof of the Comparison Test, using the Cauchy Criterion for sequences (not series). (b) Each partial product is a natural number; the infinite product is not convergent. (c) “If”: use $1+x \leq e^x$. “Only if”: use $(1+x)(1+y) \geq 1+x+y$. None of the actual estimates, the explicit formula, or the unboundedness argument in (b) is carried out.

Solution

Method 1: Direct Cauchy-sequence estimates for partial products

Part (a). (*Scratch work – not part of the written answer.*) Since every factor is ≥ 1 , both $\{p_n\}$ and $\{q_n\}$ (partial products of $\prod b_n, \prod c_n$) are increasing, and $p_n \leq q_n$ for every n ; bound the increment $p_n - p_m$ by the corresponding increment $q_n - q_m$, then transfer Cauchy-ness of $\{q_n\}$ to $\{p_n\}$ directly (no Monotone Convergence Theorem).

Proof. Let $p_n := b_1 b_2 \cdots b_n$ and $q_n := c_1 c_2 \cdots c_n$ for $n \in \mathbb{N}$.

Monotonicity and a global comparison. Since $b_k \geq 1$ for every k , $p_{n+1} = p_n \cdot b_{n+1} \geq p_n \cdot 1 = p_n$ (as $p_n \geq 1 > 0$, being a product of factors ≥ 1), so $\{p_n\}$ is increasing; likewise $\{q_n\}$ is increasing. By induction on n , using $1 \leq b_k \leq c_k$ for every k (and that all partial

products up to index $k-1$ are positive, so multiplying preserves the inequality direction), $p_n \leq q_n$ for every $n \in \mathbb{N}$.

The key increment inequality. Fix $n, m \in \mathbb{N}$ with $n > m$. Since $1 \leq b_k \leq c_k$ for every k , multiplying these inequalities over $k = m+1, \dots, n$ (all factors positive) gives $1 \leq b_{m+1} \cdots b_n \leq c_{m+1} \cdots c_n$, hence $0 \leq b_{m+1} \cdots b_n - 1 \leq c_{m+1} \cdots c_n - 1$. Using $p_m \leq q_m$ and $p_m \geq 0$,

$$0 \leq p_n - p_m = p_m(b_{m+1} \cdots b_n - 1) \leq q_m(b_{m+1} \cdots b_n - 1) \leq q_m(c_{m+1} \cdots c_n - 1) = q_n - q_m.$$

Thus $0 \leq p_n - p_m \leq q_n - q_m$ whenever $n > m$; by the symmetry of the absolute value, this also gives $|p_n - p_m| \leq |q_n - q_m|$ for all $n, m \in \mathbb{N}$ (not just $n > m$).

Conclusion. Since $\prod c_n$ is convergent, $\{q_n\}$ is convergent, hence Cauchy (Cauchy Criterion for sequences). Let $\varepsilon > 0$. There is $N \in \mathbb{N}$ such that for all $n, m \in \mathbb{N}$, $n, m \geq N \implies |q_n - q_m| < \varepsilon$. For such n, m , $|p_n - p_m| \leq |q_n - q_m| < \varepsilon$ as well. Hence $\{p_n\}$ is a Cauchy sequence, so by the Cauchy Criterion (for sequences), $\{p_n\}$ is convergent; that is, $\prod b_n$ is convergent.

$$(a) \ 1 \leq b_n \leq c_n \ \forall n, \ \prod c_n \text{ convergent} \implies \prod b_n \text{ convergent.}$$

□

Part (b). (*Scratch work – not part of the written answer.*) Telescope the product of $\frac{k+1}{k}$ terms.

Solution. For $a_n = 1/n$, and every $n \in \mathbb{N}$,

$$p_n = \prod_{k=1}^n \left(1 + \frac{1}{k}\right) = \prod_{k=1}^n \frac{k+1}{k} = \frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n} = n+1,$$

since the product telescopes (each numerator cancels the next denominator). *Check:* $p_1 = 1+1 = 2 = 1+1$; $p_2 = 2 \cdot (1+\frac{1}{2}) = 2 \cdot \frac{3}{2} = 3 = 2+1$; $p_3 = 3 \cdot (1+\frac{1}{3}) = 3 \cdot \frac{4}{3} = 4 = 3+1$ – consistent with the formula $p_n = n+1$. Since $p_n = n+1$ for every $n \in \mathbb{N}$, given any $K \in \mathbb{R}$, the Archimedean Property gives $n \in \mathbb{N}$ with $n > K-1$, so $p_n = n+1 > K$; hence $\{p_n\}$ is unbounded above. By the Boundedness of Convergent Sequences (contrapositive), an unbounded sequence cannot converge, so $\{p_n\}$ diverges.

$$(b) \ p_n = n+1 \text{ for all } n \in \mathbb{N}; \text{ the infinite product } \prod (1 + 1/n) \text{ diverges } (p_n \rightarrow \infty).$$

Part (c). (*Scratch work – not part of the written answer.*) “If”: the exponential function is not course material (Lectures 1–17), so the usual bound $1+x \leq e^x$ cannot be used.

Instead, prove by induction the purely algebraic bound $\prod_{i=1}^r (1+x_i) \leq \frac{1}{1 - \sum_{i=1}^r x_i}$ for nonnegative x_i with $\sum_{i=1}^r x_i < 1$; combined with the Cauchy Criterion for Series applied to $\sum a_n$ (to force tails below $\frac{1}{2}$), this bounds p_m globally and controls $p_n - p_m$, exactly mirroring the role $e^{s_n - s_m}$ would have played. “Only if”: extend $(1+x)(1+y) \geq 1+x+y$ by induction to get $p_n/p_m \geq 1 + (s_n - s_m)$, and note $p_m \geq 1$ makes this comparison very direct.

Proof (“if”: $\sum a_n$ **convergent** $\implies \prod(1 + a_n)$ **convergent**). Let $s_n := a_1 + a_2 + \dots + a_n$ be the n -th partial sum of $\sum a_n$, and suppose $\sum a_n$ is convergent. Since the exponential function is not available in this course, we first establish a purely algebraic replacement for the bound $\prod(1 + x_i) \leq e^{\sum x_i}$.

Step 1 (an elementary product bound). We claim that for every $r \in \mathbb{N}$ and all non-negative reals $x_1, \dots, x_r$ with $S_r := \sum_{i=1}^r x_i < 1$,

$$\prod_{i=1}^r (1 + x_i) \leq \frac{1}{1 - S_r}.$$

We prove this by induction on r . For $r = 1$, since $x_1 \geq 0$, $(1 + x_1)(1 - x_1) = 1 - x_1^2 \leq 1$; as $1 - x_1 > 0$, dividing both sides by $1 - x_1$ gives $1 + x_1 \leq \frac{1}{1 - x_1}$, which is the claim for $r = 1$. Assume the claim holds for some $r \in \mathbb{N}$, and let $x_1, \dots, x_{r+1} \geq 0$ satisfy $S_{r+1} := S_r + x_{r+1} < 1$ (so also $S_r < 1$). By the inductive hypothesis and $1 + x_{r+1} > 0$,

$$\prod_{i=1}^{r+1} (1 + x_i) = \left(\prod_{i=1}^r (1 + x_i) \right) (1 + x_{r+1}) \leq \frac{1 + x_{r+1}}{1 - S_r}.$$

Since $1 - S_r > 0$ and $1 - S_{r+1} > 0$, and

$$(1 - S_r) - (1 + x_{r+1})(1 - S_{r+1}) = (1 - S_r) - (1 - S_r - x_{r+1}S_{r+1}) = x_{r+1}S_{r+1} \geq 0,$$

(using $S_{r+1} = S_r + x_{r+1}$), we get $(1 + x_{r+1})(1 - S_{r+1}) \leq 1 - S_r$, so, dividing by the positive quantities $1 - S_r$ and $1 - S_{r+1}$,

$$\frac{1 + x_{r+1}}{1 - S_r} \leq \frac{1}{1 - S_{r+1}}.$$

Combining the last two displays, $\prod_{i=1}^{r+1} (1 + x_i) \leq \frac{1}{1 - S_{r+1}}$, completing the induction.

This proves Step 1.

Step 2 (a global bound on the partial products). Since $\sum a_n$ is convergent, the Cauchy Criterion for Series (applied with $\varepsilon = \frac{1}{2}$) gives $N_0 \in \mathbb{N}$ such that for all $j, k \in \mathbb{N}$ with $k > j \geq N_0$,

$$0 \leq s_k - s_j = a_{j+1} + a_{j+2} + \dots + a_k < \frac{1}{2}$$

(the terms $a_n \geq 0$ let us drop the absolute value). Let $K := 2p_{N_0}$, a fixed positive real number. For every $n > N_0$, applying Step 1 with $r = n - N_0$ and $x_i := a_{N_0+i}$ (so $S_r = s_n - s_{N_0} < \frac{1}{2} < 1$ by the choice of N_0),

$$\frac{p_n}{p_{N_0}} = \prod_{k=N_0+1}^n (1 + a_k) \leq \frac{1}{1 - (s_n - s_{N_0})} < \frac{1}{1 - 1/2} = 2,$$

so $p_n < 2p_{N_0} = K$. Since also $p_{N_0} < K$, we conclude $p_n < K$ for every $n \geq N_0$.

Step 3 (bounding the increments). Fix $n, m \in \mathbb{N}$ with $n > m \geq N_0$. Applying Step 1 with $r = n - m$ and $x_i := a_{m+i}$ (so $S_r = s_n - s_m < \frac{1}{2} < 1$, since $n > m \geq N_0$),

$$\frac{p_n}{p_m} = \prod_{k=m+1}^n (1 + a_k) \leq \frac{1}{1 - (s_n - s_m)}.$$

Since $\{p_n\}$ is increasing (each factor $1 + a_k \geq 1$, as in Part (a)), $p_n \geq p_m > 0$, so

$$0 \leq p_n - p_m = p_m \left(\frac{p_n}{p_m} - 1 \right) \leq p_m \cdot \frac{s_n - s_m}{1 - (s_n - s_m)}.$$

Since $s_n - s_m < \frac{1}{2}$, $1 - (s_n - s_m) > \frac{1}{2}$, so $\frac{s_n - s_m}{1 - (s_n - s_m)} < 2(s_n - s_m)$; together with $p_m < K$ (Step 2, as $m \geq N_0$), this gives

$$0 \leq p_n - p_m < 2K(s_n - s_m) \quad \text{for all } n > m \geq N_0.$$

Step 4 (Cauchy estimate). Let $\varepsilon > 0$ be given, and set $\varepsilon' := \min \left\{ \frac{\varepsilon}{2K+1}, \frac{1}{2} \right\} > 0$.

Since $\sum a_n$ is convergent, the Cauchy Criterion for Series gives $N'_1 \in \mathbb{N}$ such that for all $n, m \in \mathbb{N}$, $n > m \geq N'_1 \implies 0 \leq s_n - s_m < \varepsilon'$. Let $N_1 := \max\{N_0, N'_1\}$. Fix $n, m \in \mathbb{N}$ with $n > m \geq N_1$; then also $n > m \geq N'_1$, so $0 \leq s_n - s_m < \varepsilon' \leq \frac{1}{2}$, and $n > m \geq N_0$, so Step 3 applies:

$$0 \leq p_n - p_m < 2K(s_n - s_m) < 2K\varepsilon' \leq 2K \cdot \frac{\varepsilon}{2K+1} < \varepsilon.$$

Thus, for every $\varepsilon > 0$, there is $N_1 \in \mathbb{N}$ such that for all $n, m \in \mathbb{N}$, $n > m \geq N_1 \implies |p_n - p_m| < \varepsilon$ (dropping the absolute value, and symmetrising in n, m , since $\{p_n\}$ is increasing). Hence $\{p_n\}$ is a Cauchy sequence, so by the Cauchy Criterion (for sequences), $\{p_n\}$ is convergent; that is, $\prod(1 + a_n)$ is convergent. $\square$

Proof (“only if”: $\prod(1 + a_n)$ convergent $\implies \sum a_n$ convergent). Suppose $\prod(1 + a_n)$ is convergent, i.e. $\{p_n\}$ is convergent. First, since $a_k \geq 0$ for every k , each factor $1 + a_k \geq 1$, so (as in part (a)) $\{p_n\}$ is increasing and $p_n \geq p_1 \cdots$; in particular $p_n \geq 1$ for every $n \in \mathbb{N}$ (empty/trivial product convention $p_0 := 1$, or directly since each factor ≥ 1).

By induction on $n - m$, using $(1 + x)(1 + y) \geq 1 + x + y$ for $x, y \geq 0$ repeatedly, for every $n > m$,

$$\prod_{k=m+1}^n (1 + a_k) \geq 1 + \sum_{k=m+1}^n a_k = 1 + (s_n - s_m).$$

(Base case $n = m + 1$ is the single factor $1 + a_{m+1} = 1 + (s_{m+1} - s_m)$, an equality. Inductive step: if $\prod_{k=m+1}^{n-1} (1 + a_k) \geq 1 + (s_{n-1} - s_m)$, multiply both sides by the positive factor $(1 + a_n)$ and apply $(1 + x)(1 + y) \geq 1 + x + y$ with $x = s_{n-1} - s_m \geq 0$, $y = a_n \geq 0$, to get $\prod_{k=m+1}^n (1 + a_k) \geq (1 + (s_{n-1} - s_m))(1 + a_n) \geq 1 + (s_{n-1} - s_m) + a_n = 1 + (s_n - s_m)$.)

Since $p_n/p_m = \prod_{k=m+1}^n (1 + a_k) \geq 1 + (s_n - s_m)$, and $p_m \geq 1 > 0$,

$$s_n - s_m \leq \frac{p_n}{p_m} - 1 = \frac{p_n - p_m}{p_m} \leq \frac{p_n - p_m}{1} = p_n - p_m,$$

using $p_m \geq 1$ to pass from dividing by p_m to dividing by 1 (both $p_n - p_m \geq 0$ and $p_m \geq 1$, so dividing a nonnegative quantity by something ≥ 1 only shrinks it). Also $s_n - s_m \geq 0$ since $a_k \geq 0$, so $|s_n - s_m| = s_n - s_m \leq p_n - p_m = |p_n - p_m|$ for $n \geq m$.

Since $\{p_n\}$ is convergent, it is Cauchy (Cauchy Criterion for sequences). Let $\varepsilon > 0$. There is $N \in \mathbb{N}$ such that for all $n, m \in \mathbb{N}$, $n > m \geq N \implies |p_n - p_m| < \varepsilon$. For such n, m , $|s_n - s_m| \leq |p_n - p_m| < \varepsilon$. Hence $\{s_n\}$ is Cauchy, so by the Cauchy Criterion for Series, $\sum a_n$ is convergent.

$$(c) \quad \prod(1 + a_n) \text{ convergent} \iff \sum a_n \text{ convergent} \quad (a_n \geq 0).$$

$\square$

Beyond the hints. The official hint’s suggested inequality for the “if” direction, $1 + x \leq e^x$, uses the exponential function, which is *not* course material in Lectures 1–17; the proof above replaces it with a purely algebraic induction ($\prod_{i=1}^r (1 + x_i) \leq 1/(1 - \sum_{i=1}^r x_i)$ for nonnegative x_i summing to less than 1) that plays exactly the same role without appealing to anything outside the syllabus. Neither direction’s estimate is carried out in the hint: for “if”, the induction proving this product bound, the global bound K on $\{p_n\}$, and the final ε – N argument are all supplied above; for “only if”, the induction extending $(1 + x)(1 + y) \geq 1 + x + y$ to $n - m$ factors and the bound $p_m \geq 1$ used to remove the division are supplied above. In part (a), the hint says to “follow the proof of the Comparison Test”; the increment inequality $0 \leq p_n - p_m \leq q_n - q_m$ derived above is the exact multiplicative analogue of the additive Comparison Test estimate from Lecture 12.

Common slip. In part (a), the Cauchy Criterion needed is the one *for sequences* (applied to $\{p_n\}, \{q_n\}$ directly), not the Cauchy Criterion *for series* – there is no series of partial sums here, only a sequence of partial products; the hint flags this explicitly, and part (c) similarly mixes both: the Cauchy Criterion for sequences bounds $\{p_n\}$, while the Cauchy Criterion for Series is used for $\{s_n\}$. In part (b), the reasoning “ p_n is always a natural number, and a convergent sequence of natural numbers would have to be eventually constant” is an alternative valid argument to the one given above (which instead shows $p_n \rightarrow \infty$ directly); either is acceptable, but merely observing “ $p_n \in \mathbb{N}$ ” without an unboundedness or eventually-constant argument does not by itself prove divergence.

Question A3 (Alternating Series Test via NIP)

Problem

The Alternating Series Test can be proved directly from the Nested Interval Property.

- (a) Given an alternating series $b_1 - b_2 + b_3 - b_4 + \cdots$ with the sequence $\{b_n\}$ decreasing and converging to 0, construct a nested sequence of closed and bounded intervals $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$, whose lengths converge to 0, such that for each $n, m \in \mathbb{N}$, $m \geq n \implies s_m \in I_n$, where s_m denotes the m -th partial sum of the alternating series.

- (b) Hence use the Nested Interval Property to deduce a sum for the alternating series.

(Tutorial 4, Additional Question 3.)

Official hint. (a) Recall that the lecture proof of the Alternating Series Test establishes that for every $N, n \in \mathbb{N}$, $n > N \implies 0 \leq b_{N+1} - b_{N+2} + \cdots + (-1)^{n-N-1}b_n \leq b_{N+1}$. (b) The Nested Interval Property gives a real number s ; show s is the sum via the definition of a limit, remembering $\{b_n\} \rightarrow 0$. No explicit interval, no nesting check, and no final ε - N argument are given.

Solution

Method 1: $I_n =$ the interval spanned by consecutive partial sums

Part (a). (*Scratch work – not part of the written answer.*) Set $s_0 := 0$ (the empty partial sum). Apply the quoted lecture inequality with $N = n - 1$ (so that “ $n > N$ ” in the quoted fact becomes “ $m \geq n$ ”, matching what we need); this rewrites, after tracking the sign $(-1)^{n-1}$, as s_m lying between s_{n-1} and s_n for every $m \geq n$. So the natural candidate is $I_n :=$ the closed interval with endpoints s_{n-1} and s_n .

Construction. For $n \in \mathbb{N}$, define

$$I_n := [\min(s_{n-1}, s_n), \max(s_{n-1}, s_n)],$$

using the convention $s_0 := 0$. We verify the three required properties.

Containment. Fix $n \in \mathbb{N}$ and apply the quoted lecture inequality with $N := n - 1 \in \mathbb{N}_0$: for every $m \in \mathbb{N}$ with $m > n - 1$ (i.e. $m \geq n$),

$$0 \leq b_n - b_{n+1} + b_{n+2} - \cdots + (-1)^{m-n}b_m \leq b_n.$$

The left-hand quantity equals $(-1)^{n-1}(s_m - s_{n-1})$: indeed, the k -th term of the quoted sum (at index $k = n, n+1, \dots, m$) is $(-1)^{k-n}b_k$, while the k -th term of $s_m - s_{n-1} = \sum_{k=n}^m (-1)^{k-1}b_k$ is $(-1)^{k-1}b_k$, and $(-1)^{k-1} = (-1)^{n-1}(-1)^{k-n}$. So the quoted inequality reads $0 \leq (-1)^{n-1}(s_m - s_{n-1}) \leq b_n$ for every $m \geq n$.

If n is odd, $(-1)^{n-1} = 1$, so $0 \leq s_m - s_{n-1} \leq b_n$ for every $m \geq n$, i.e. $s_{n-1} \leq s_m \leq s_{n-1} + b_n$. Since $s_n = s_{n-1} + (-1)^{n-1}b_n = s_{n-1} + b_n$ (as n is odd), this says $s_m \in [s_{n-1}, s_n] = I_n$ for every $m \geq n$.

If n is even, $(-1)^{n-1} = -1$, so $0 \leq -(s_m - s_{n-1}) \leq b_n$ for every $m \geq n$, i.e. $s_{n-1} - b_n \leq s_m \leq s_{n-1}$. Since $s_n = s_{n-1} + (-1)^{n-1}b_n = s_{n-1} - b_n$ (as n is even), this says $s_m \in [s_n, s_{n-1}] = I_n$ for every $m \geq n$.

In both cases, $m \geq n \implies s_m \in I_n$, as required.

Nesting. Fix $n \in \mathbb{N}$. Taking $m = n$ and $m = n + 1$ (both $\geq n$) in the containment property just proved, $s_n \in I_n$ and $s_{n+1} \in I_n$. Since I_n is a closed interval containing both s_n and s_{n+1} , and $I_{n+1} = [\min(s_n, s_{n+1}), \max(s_n, s_{n+1})]$ is by definition the *smallest* closed interval containing both s_n and s_{n+1} , $I_{n+1} \subseteq I_n$. As this holds for every $n \in \mathbb{N}$, $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$.

Lengths $\rightarrow 0$. First, $b_n \geq 0$ for every $n \in \mathbb{N}$: since $\{b_n\}$ is decreasing, $b_n \geq b_k$ for every $k \geq n$, and letting $k \rightarrow \infty$, the Order Limit Theorem gives $b_n \geq \lim_{k \rightarrow \infty} b_k = 0$. Hence, for every $n \in \mathbb{N}$, the length of I_n is $|I_n| = |s_n - s_{n-1}| = |(-1)^{n-1}b_n| = b_n$. Since $\{b_n\}$ converges to 0 by hypothesis, $|I_n| = b_n \rightarrow 0$.

Each I_n is closed and bounded (a closed interval with finite endpoints), so $\{I_n\}$ is a nested sequence of closed and bounded intervals with lengths converging to 0, and $m \geq n \implies s_m \in I_n$ for every $n, m \in \mathbb{N}$, as required.

Part (b). (*Scratch work – not part of the written answer.*) Nested Interval Property gives a point in every I_n ; lengths $\rightarrow 0$ make this point unique, and then the same length bound gives $s_m \rightarrow s$ directly.

Proof. By the Nested Interval Property, applied to $I_1 \supseteq I_2 \supseteq \dots$, there is $s \in \mathbb{R}$ with $s \in I_n$ for every $n \in \mathbb{N}$.

Uniqueness of s (used only to justify calling s “the” limit). If $s' \in \mathbb{R}$ also satisfies $s' \in I_n$ for every n , then for every $n \in \mathbb{N}$, both $s, s' \in I_n$, so $|s - s'| \leq |I_n| = b_n$. Since $b_n \rightarrow 0$, the Order Limit Theorem (letting $n \rightarrow \infty$ in $0 \leq |s - s'| \leq b_n$) gives $|s - s'| \leq 0$, so $s = s'$.

$s_m \rightarrow s$. Let $\varepsilon > 0$. Since $\{b_n\} \rightarrow 0$, there is $N \in \mathbb{N}$ such that $b_N < \varepsilon$ (taking, e.g., N from the definition of convergence of $\{b_n\}$ to 0 applied at index N itself). For every $m \geq N$: by part (a) (containment with $n = N$), $s_m \in I_N$; and since $s \in I_k$ for every $k \in \mathbb{N}$, in particular $s \in I_N$. As s_m and s both lie in the interval I_N of length b_N ,

$$|s_m - s| \leq |I_N| = b_N < \varepsilon.$$

Thus, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that $m \geq N \implies |s_m - s| < \varepsilon$; this is precisely the definition of $\{s_m\} \rightarrow s$. Hence the alternating series $\sum (-1)^{n-1}b_n = b_1 - b_2 + b_3 - \dots$ converges, with sum s , where s is the unique point of $\bigcap_{n=1}^{\infty} I_n$.

$$b_1 - b_2 + b_3 - \dots \text{ converges to the unique } s \in \bigcap_{n=1}^{\infty} I_n.$$

□

Beyond the hints. The official hint quotes the lecture’s inequality and says to use the Nested Interval Property, but constructs no interval, checks no nesting, and never performs the final ε – N argument. The explicit choice $I_n = [\min(s_{n-1}, s_n), \max(s_{n-1}, s_n)]$, the sign bookkeeping converting the quoted lecture inequality into containment, the nesting proof (via “smallest interval containing two already-contained points”), the length computation $|I_n| = b_n$, the uniqueness argument for s , and the closing ε – N proof that $s_m \rightarrow s$ are all supplied here.

Common slip. A tempting but wrong shortcut in part (b) is to declare “ $s = \lim s_{2n} = \lim s_{2n-1}$ ” without justification – this is exactly what the proof of the Alternating Series Test via the Monotone Convergence Theorem does (splitting into odd/even subsequences), which is precisely the route this construction is designed to *avoid*; the Nested

Interval Property argument above never needs to separately show the odd and even sub-sequences converge to a common limit, since containment in the shrinking intervals I_n gives convergence of the *whole* sequence $\{s_m\}$ directly.

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Method Map

Q.	Core topic	Methods included
1	Convergence of three series	Comparison Test against a geometric series and $\sum 1/n^2$; Absolute Convergence Test; splitting an irregular alternating-type series into a divergent and a convergent piece, closed with Subsequential Limits.
2	True/false with examples	Explicit convergent/divergent examples verified by direct computation; a proof via the Algebraic Limit Theorem for Series (part (c)).
3	Proof or counterexample	Comparison Test via a global bound from Boundedness of Convergent Sequences (a); a squared-alternating-series counterexample (b); proof by contrapositive using Terms of Convergent Series and Comparison Test (c).
4	Cauchy partial sums, definitions only	Direct ε - N estimate re-deriving the Comparison-Test inequality without naming the banned theorems.
5	Regrouping preserves the sum	Identifying partial sums of the regrouped series as a subsequence of the original partial sums; Subsequential Limits.
A1	Abel's Test	Abel's Lemma (summation by parts) proved from scratch; a tail-shifted version to avoid needing $b_n \rightarrow 0$; Cauchy Criterion for Series.
A2	Infinite products	Cauchy Criterion for sequences applied to partial products; a telescoping explicit formula; an elementary algebraic product bound (replacing the out-of-scope $1 + x \leq e^x$) together with the linear bracketing inequality $(1 + x)(1 + y) \geq 1 + x + y$ for the product-series equivalence.
A3	Alternating Series Test via NIP	Explicit nested intervals from consecutive partial sums; Nested Interval Property; direct ε - N convergence proof.

Technique Summary

- The Cauchy Criterion for Series is the master tool of Lectures 10–12.** The Comparison Test, the Absolute Convergence Test, and the Alternating Series Test are all proved by bounding a tail sum $|a_{m+1} + \cdots + a_n|$ of the target series by a tail sum of a reference series or quantity that can be made small; Question 4 and Additional Question 1 show this estimate can always be re-derived by hand when a named test is banned.
- A global bound beats an eventual bound.** Boundedness of Convergent Sequences gives a constant M (or b_1) that bounds a sequence for *all* indices at once, which is what allows a Comparison-Test-style inequality to hold for every n , not just eventually (Questions 1, 3, 4, Additional Question 1).

3. **Absolute convergence controls related series.** If $\sum |a_n|$ converges, then so does any series whose terms are dominated by a constant multiple of $|a_n|$ (Question 3(a), (c)); conditional convergence is fragile and easily destroyed by multiplying by another oscillating, non-decaying factor (Question 2(b), Question 3(b)).
4. **Partial sums of a regrouped or product-summed series are often literally a subsequence.** Recognising this (Question 5, and the increasing-partial-product argument of Additional Question 2) reduces the problem to the Subsequential Limits theorem.
5. **The harmonic series and $\sum 1/n^2$ are the two workhorse comparison series** for this entire tutorial: the first proves divergence (Questions 1(c), 2(a), 2(b), 2(d), 3(b)), the second proves convergence (Questions 1(b), 1(c), 2(a), 3(a), 3(c)).

Final Answers

1. (a) Convergent. (b) Convergent. (c) Divergent.
2. (a) True. (b) True. (c) False. (d) True.
3. (a) True. (b) False. (c) True.
4. Proved directly from the ε - N definitions of Cauchy partial sums (no named test used).
5. Proved via $\sigma_k = s_{n_k}$ and the Subsequential Limits theorem.
- A1. Proved via Abel's Lemma and a tail-shifted Cauchy Criterion estimate (Abel's Test).
- A2. (a) Proved via the Cauchy Criterion for sequences. (b) $p_n = n + 1$; divergent. (c) $\prod (1 + a_n)$ convergent iff $\sum a_n$ convergent.
- A3. $I_n = [\min(s_{n-1}, s_n), \max(s_{n-1}, s_n)]$; the alternating series converges, with sum the unique point of $\bigcap_n I_n$ given by the Nested Interval Property.